
FECAP

RELATÓRIO DE CONFIGURAÇÃO DE AMBIENTE DE COMPUTAÇÃO EM NUVEM

Projeto Integrador – 5º Semestre

Sistemas Operacionais e Computação em Nuvem

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1. Introdução

O presente relatório tem como objetivo documentar o processo de configuração de uma máquina virtual na plataforma de nuvem Amazon Web Services (AWS), utilizando o serviço Amazon EC2 (Elastic Compute Cloud). A atividade integra a disciplina de Projeto Integrador do 5º semestre do curso de Ciência da Computação da FECAP, com foco na prática de administração de servidores Linux em ambiente de nuvem.

A configuração envolveu a criação de uma instância EC2 com sistema operacional Ubuntu Server 24.04 LTS, a instalação das ferramentas necessárias (Git, Python 3 e PostgreSQL), a clonagem de um repositório GitHub contendo um projeto de backend desenvolvido com FastAPI, e a execução da aplicação com banco de dados em nuvem.

2. Descrição do Ambiente

As especificações do ambiente configurado são detalhadas abaixo:

- **Plataforma de Nuvem:** Amazon Web Services (AWS)
- **Serviço Utilizado:** Amazon EC2 (Elastic Compute Cloud)
- **Região:** us-east-2 (Ohio, Estados Unidos)
- **Sistema Operacional:** Ubuntu Server 24.04 LTS (HVM), EBS General Purpose SSD
- **Tipo de Instância:** t3.micro (2 vCPUs, 1 GiB de memória RAM)
- **Armazenamento:** 8 GiB (EBS – SSD de uso geral)
- **ID da Instância:** i-026d329df00285aa0 (fecap-instance-ubuntu)
- **IP Público IPv4:** 3.17.60.86
- **Grupo de Segurança:** sg-02d8404b526e17915 (launch-wizard-1)
- **Método de Conexão:** EC2 Instance Connect (via navegador)
- **Linguagem de Backend:** Python 3 com FastAPI e Uvicorn
- **Banco de Dados:** PostgreSQL (instalado localmente na instância)

3. Passo a Passo da Configuração

3.1 Acesso à Instância EC2

Após criar a instância EC2 com Ubuntu 24.04 no console da AWS, o acesso foi realizado pelo recurso nativo do AWS chamado EC2 Instance Connect, que permite abrir um terminal Linux diretamente no navegador sem a necessidade de configurar chaves SSH localmente. Para isso, basta navegar até EC2 > Instâncias, selecionar a instância desejada, clicar em Conectar e escolher a aba Conexão de instância do EC2.

3.2 Atualização do Sistema

O primeiro passo após conectar à instância foi garantir que todos os pacotes do sistema operacional estivessem atualizados. Isso evita conflitos de versões e garante acesso às versões mais recentes dos repositórios:

```
sudo apt update
sudo apt upgrade -y
```

O parâmetro `-y` confirma automaticamente todas as perguntas de instalação, evitando interrupções manuais durante o processo.

3.3 Instalação do Git

O Git foi instalado para permitir a clonagem de repositórios diretamente na instância:

```
sudo apt install git -y
git --version
```

O segundo comando verifica se a instalação foi bem-sucedida, exibindo a versão instalada (ex: git version 2.43.0).

3.4 Instalação do Python 3

O Python 3 foi instalado juntamente com o gerenciador de pacotes pip e o módulo venv, necessário para criar ambientes virtuais isolados:

```
sudo apt install python3 python3-pip python3-venv -y
python3 --version
pip3 --version
```

3.5 Instalação do PostgreSQL

O PostgreSQL foi instalado diretamente na instância para servir como banco de dados relacional da aplicação:

```
sudo apt install postgresql postgresql-contrib -y
sudo systemctl start postgresql
sudo systemctl enable postgresql
sudo systemctl status postgresql
```

O comando `enable` garante que o serviço seja iniciado automaticamente a cada reinicialização da instância. O status deve exibir `active (running)` para confirmar o funcionamento correto.

3.6 Clonagem do Repositório

Uma pasta de trabalho foi criada para organizar os projetos, e o repositório foi clonado diretamente do GitHub:

```
mkdir ~/fecap-projects
cd fecap-projects/
git clone https://github.com/2026-1-NCC5/Projeto10.git
cd Projeto10/
```

O clone foi realizado com sucesso, transferindo 208.44 MiB de dados do repositório contendo 1141 objetos e 1879 arquivos.

3.7 Configuração do Ambiente Virtual e Dependências

Para isolar as dependências do projeto, um ambiente virtual Python foi criado dentro do diretório do backend:

```
cd src/Entrega1/backend/backend-api/
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

O arquivo requirements.txt continha as seguintes dependências principais do projeto:

- FastAPI (framework web)
- Uvicorn (servidor ASGI)
- SQLAlchemy (ORM para banco de dados)
- Alembic (migrations de banco de dados)
- Psycopg2-binary (driver PostgreSQL para Python)
- Python-dotenv (gerenciamento de variáveis de ambiente)
- Python-jose e Passlib (autenticação JWT e criptografia)
- Pydantic (validação de dados)

3.8 Configuração do Banco de Dados

O banco de dados PostgreSQL foi configurado com a definição de senha do usuário padrão e a criação do banco de dados da aplicação:

```
sudo -u postgres psql -c "ALTER USER postgres PASSWORD 'postgres';"
sudo -u postgres createdb empathic_leaders
```

Em seguida, as migrations foram executadas com o Alembic para criar as tabelas definidas nos modelos do SQLAlchemy:

```
alembic upgrade head
```

As variáveis de ambiente foram configuradas em um arquivo .env no diretório do backend, contendo informações de conexão com o banco de dados, chave secreta JWT e outras configurações da aplicação.

3.9 Execução da Aplicação

Com todas as dependências instaladas e o banco configurado, a aplicação foi iniciada com o servidor Uvicorn:

```
uvicorn main:app --host 0.0.0.0 --port 8001
```

A aplicação subiu com sucesso e respondeu a requisições HTTP, conforme evidenciado pelos logs no terminal:

- `POST /api/auth/register HTTP/1.1 - 200 OK`
- `OPTIONS /api/users/me/role HTTP/1.1 - 200 OK`
- `PUT /api/users/me/role HTTP/1.1 - 200 OK`

Para validar os dados persistidos, foi executada uma consulta SQL diretamente no banco de dados para confirmar que o registro do usuário havia sido salvo corretamente:

```
sudo -u postgres psql -d empathic_leaders -c "SELECT name, email, role  
FROM users;"
```

4. Dificuldades Encontradas e Soluções

Durante o processo de configuração, algumas dificuldades foram encontradas e solucionadas:

4.1 Falta de experiência prévia com instâncias EC2

Por ser a primeira vez utilizando o AWS EC2, o processo de criação e conexão à instância demandou atenção extra. A criação do par de chaves RSA no formato .pem, a definição do grupo de segurança e a compreensão da diferença entre IP público e privado foram pontos que exigiram estudo. A solução foi utilizar o método de conexão EC2 Instance Connect, que dispensa o uso de cliente SSH local.

4.2 Configuração do arquivo .env

O arquivo .env com as variáveis de ambiente não é versionado no repositório (por boas práticas de segurança), então foi necessário criá-lo manualmente na instância. As variáveis incluíram a string de conexão com o PostgreSQL, a chave secreta para geração de tokens JWT e as configurações de algoritmo de criptografia.

4.3 Execução das migrations com Alembic

A execução do comando alembic upgrade head exigiu que o ambiente virtual estivesse ativo e que o arquivo alembic.ini estivesse apontando corretamente para a URL do banco de dados. Após ajustar a DATABASE_URL no arquivo .env e garantir que o Alembic estava lendo as variáveis de ambiente, as tabelas foram criadas com sucesso.

5. Capturas de Tela

As capturas de tela a seguir documentam as principais etapas realizadas durante a configuração do ambiente. As imagens estão organizadas em ordem cronológica de execução:

The image displays two screenshots of the AWS Management Console interface.

The top screenshot shows the **Console Home** dashboard. It includes sections for **Recently visited** (No recently visited services), **Applications** (0 applications), **Welcome to AWS** (Getting started with AWS, Training and certification), **AWS Health** (Open issues: 0, Scheduled changes: 0, Other notifications: 0), and **Cost and usage** (Data unavailable). The dashboard also features a search bar, a region selector (United States (Ohio)), and a user profile (SauloPJ).

The bottom screenshot shows the **Launch an instance** page. It includes a **Summary** section on the right with the following details:

- Number of instances:** 1
- Software Image (AMI):** Amazon Linux 2023 AMI 2023.10...read more
- Virtual server type (instance type):** t3.micro
- Firewall (security group):** New security group
- Storage (volumes):** 1 volume(s) - 8 GiB

The main content area shows the **Application and OS Images (Amazon Machine Image)** section, which includes a search bar and a grid of operating system images (Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, Debian). The **Quick Start** section is also visible, showing the **Amazon Machine Image (AMI)** selection process.

aws [Search] [Alt+S] United States (Ohio) SauloPJ (8713-4688-4445) SauloPJ

EC2 > Instances > Launch an instance

Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Canonical, Ubuntu, 24.04, amd64 noble image

Architecture 64-bit (x86) **AMI ID** ami-07062e2a343acc423 **Publish D** 2026-03-

▼ Instance type Info | Get advice
Instance type
t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0193 USD per Hour
On-Demand Windows base pricing: 0.0198 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux
Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure
Key pair name - required
Select

▼ Network settings Info Edit

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
Enter key pair name
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type
☒ RSA RSA encrypted private and public key pair
☐ ED25519 ED25519 encrypted private and public key pair

Private key file format
☒ .pem For use with OpenSSH
☐ .ppk For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

aws [Search] [Alt+S] United States (Ohio) SauloPJ (8713-4688-4445) SauloPJ

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-026d529df00285aa0)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
[Manage detailed monitoring](#)

Create Load Balancer
Create an application, network gateway or classic Elastic Load Balancer
[Create Load Balancer](#)

Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
[Create AWS budget](#)

Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
[Manage CloudWatch alarms](#)

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa0&...
[Alt+S]
United States (Ohio)
SaudoPJ [8713-4688-4443]
SaudoPJ

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Mar 22 17:28:25 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.4% of 6.71GB Processes:      113
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.40.145

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-40-145:~$

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145
```

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa0&...
[Alt+S]
United States (Ohio)
SaudoPJ [8713-4688-4443]
SaudoPJ

ubuntu@ip-172-31-40-145:~$ sudo apt update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1544 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1841 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [248 kB]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:17 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:19 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [16.7 kB]
Get:21 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1568 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [977 kB]
Get:23 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [320 kB]
Get:24 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:25 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.1 kB]
Get:26 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2825 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [219 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:29 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [26.8 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2689 kB]
Get:31 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [655 kB]
Get:32 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:33 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:34 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7276 B]
Get:35 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [949 B]
Get:36 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:37 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [625 kB]

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145
```



```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06...
AWS
Search [Alt+S]
United States (Ohio)
SauloPJ [8713-4688-4443]
SauloPJ

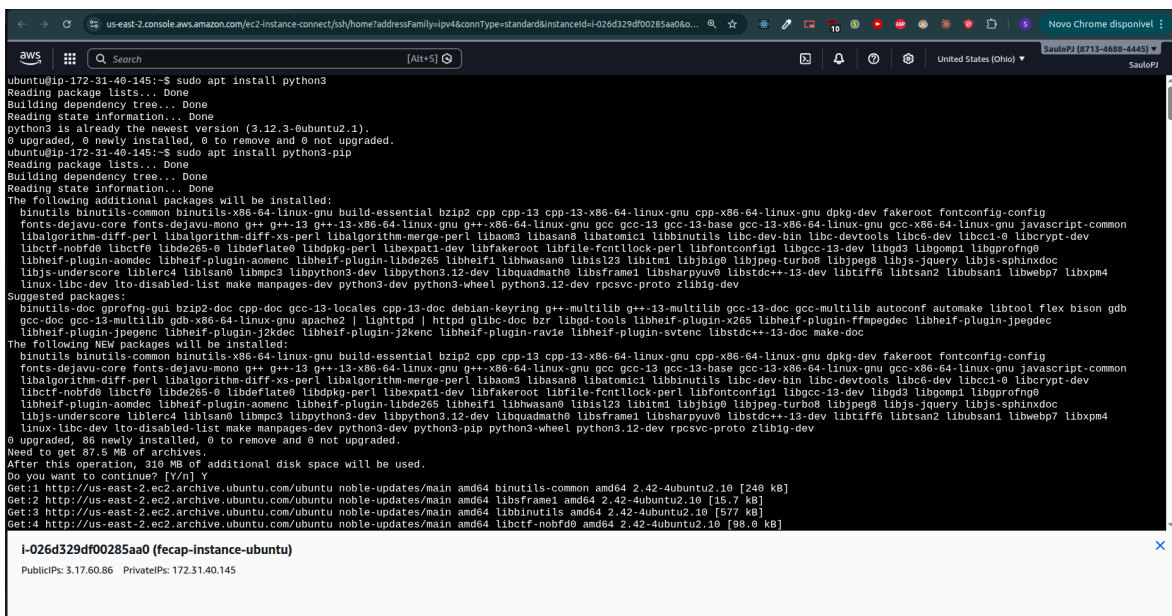
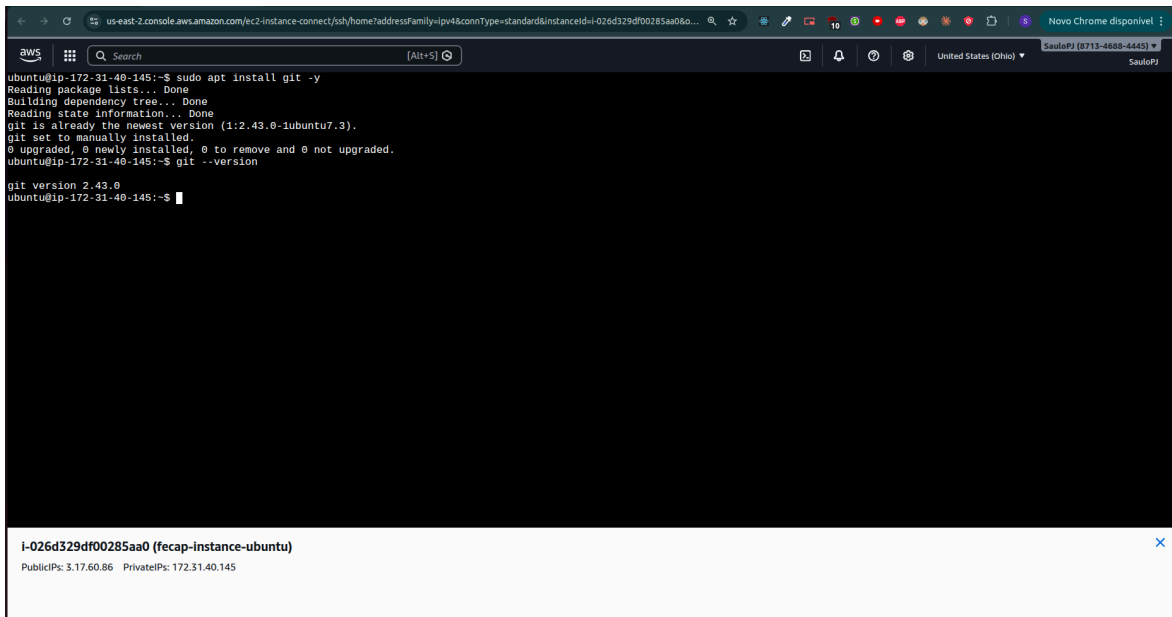
ubuntu@ip-172-31-40-145:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-aws-6.17.0-headers-6.17.0-1009 linux-aws-6.17.0-1009 linux-headers-6.17.0-1009-aws linux-image-6.17.0-1009-aws linux-modules-6.17.0-1009-aws
  linux-tools-6.17.0-1009-aws
The following packages will be upgraded:
  coreutils libssh-4 linux-aws linux-headers-aws linux-image-aws python3-cryptography snappy vim vim-common vim-runtime vim-tiny xxd
12 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
11 standard LTS security updates
Need to get 125 MB of archives.
After this operation, 197 MB of additional disk space will be used.
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils amd64 9.4-3ubuntu6.2 [1412 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim amd64 2:9.1.0016-1ubuntu7.10 [1882 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-common all 2:9.1.0016-1ubuntu7.10 [387 kB]
Get:4 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-tiny amd64 2:9.1.0016-1ubuntu7.10 [804 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-runtime all 2:9.1.0016-1ubuntu7.10 [7262 kB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 xxd amd64 2:9.1.0016-1ubuntu7.10 [54.3 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh-4 amd64 0.10.6-2ubuntu0.4 [190 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-6.17.0-1009-aws amd64 6.17.0-1009.9-24.04.2 [39.9 MB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-6.17.0-1009-aws amd64 6.17.0-1009.9-24.04.2 [15.0 MB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-aws amd64 6.17.0-1009.9-24.04.2 [1736 B]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-aws amd64 6.17.0-1009.9-24.04.2 [2324 B]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-aws-6.17.0-headers-6.17.0-1009 all 6.17.0-1009.9-24.04.2 [14.5 MB]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.17.0-1009-aws amd64 6.17.0-1009.9-24.04.2 [3720 kB]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-aws amd64 6.17.0-1009.9-24.04.2 [2240 B]
Get:15 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-aws-6.17.0-tools-6.17.0-1009 amd64 6.17.0-1009.9-24.04.2 [3491 kB]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.17.0-1009-aws amd64 6.17.0-1009.9-24.04.2 [1634 B]
Get:17 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-cryptography amd64 41.0.7-4ubuntu0.4 [615 kB]
Get:18 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 snappy amd64 2.73-5ubuntu24.04.2 [34.6 MB]
Fetched 125 MB in 1s (93.3 MB/s)
(Reading database ... 72566 files and directories currently installed.)
Preparing to unpack .../coreutils_9.4-3ubuntu6.2_amd64.deb ...
Unpacking coreutils (9.4-3ubuntu6.2) over (9.4-3ubuntu6.1) ...
Setting up coreutils (9.4-3ubuntu6.2) ...
(Reading database ... 72566 files and directories currently installed.)

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145
```

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06...
AWS
Search [Alt+S]
United States (Ohio)
SauloPJ [8713-4688-4443]
SauloPJ

ubuntu@ip-172-31-40-145:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-40-145:~$
```

```
i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145
```



```
aws
Search [Alt+S]
United States (Ohio)
SauloPJ (8713-4688-4443)
SauloPJ

ubuntu@ip-172-31-40-145:~$ sudo apt install python3-venv -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pip-whl python3-setuptools-whl python3.12-venv
The following NEW packages will be installed:
  python3-pip-whl python3-setuptools-whl python3-venv python3.12-venv
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 2430 kB of archives.
After this operation, 2783 kB of additional disk space will be used.
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1707 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.12 [5666 B]
Get:4 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-venv amd64 3.12.3-0ubuntu2.1 [1032 B]
Fetched 2430 kB in 0s (40.6 kB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 112259 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
Unpacking python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Selecting previously unselected package python3.12-venv.
Preparing to unpack .../python3.12-venv_3.12.3-1ubuntu0.12_amd64.deb ...
Unpacking python3.12-venv (3.12.3-1ubuntu0.12) ...
Selecting previously unselected package python3-venv.
Preparing to unpack .../python3-venv_3.12.3-0ubuntu2.1_amd64.deb ...
Unpacking python3-venv (3.12.3-0ubuntu2.1) ...
Setting up python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.12) ...
Setting up python3-venv (3.12.3-0ubuntu2.1) ...
Scanning processes...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145
```

```
aws
Search [Alt+S]
United States (Ohio)
SauloPJ (8713-4688-4443)
SauloPJ

ubuntu@ip-172-31-40-145:~$ python3 --version
Python 3.12.3
ubuntu@ip-172-31-40-145:~$ pip3 --version
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
ubuntu@ip-172-31-40-145:~$
```

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06a...
aws
Search [Alt+S]
United States (Ohio)
SauloPJ (8713-4688-4445)
SauloPJ

ubuntu@ip-172-31-40-145:~$ sudo apt install postgresql postgresql-contrib -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm17t64 libpq5 libtypes-serialiser-perl postgresql-16 postgresql-client-16 postgresql-client-common postgresql-common ssl-cert
Suggested packages:
  postgresql-doc postgresql-doc-16
The following NEW packages will be installed:
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm17t64 libpq5 libtypes-serialiser-perl postgresql postgresql-16 postgresql-client-16 postgresql-client-common
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
Need to get 43.6 MB of archives.
After this operation, 175 MB of additional disk space will be used.
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjson-perl all 4.10000-1 [81.9 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-client-common all 257build1.1 [36.4 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Get:4 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-common all 257build1.1 [161 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libcommon-sense-perl amd64 3.76-2build1 [28.4 kB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libtypes-serialiser-perl all 1.01-1 [11.6 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libjson-xs-perl amd64 4.040-0ubuntu0.24.04.1 [83.7 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libllvm17t64 amd64 1:17.0.6-9ubuntu1 [26.2 MB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.13-0ubuntu0.24.04.1 [148 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-client-16 amd64 16.13-0ubuntu0.24.04.1 [1298 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-16 amd64 16.13-0ubuntu0.24.04.1 [15.6 MB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql all 16+257build1.1 [11.6 kB]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 postgresql-contrib all 16+257build1.1 [11.6 kB]
Fetched 43.6 MB in 8s (50.9 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libjson-perl.
(Reading database ... 112276 files and directories currently installed.)
Preparing to unpack .../00-libjson-perl_4.10000-1_all.deb ...
Unpacking libjson-perl (4.10000-1) ...
Selecting previously unselected package postgresql-client-common.
Preparing to unpack .../01-postgresql-client-common_257build1.1_all.deb ...
Unpacking postgresql-client-common (257build1.1) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../02-ssl-cert_1.1.2ubuntu1_all.deb ...
```

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06a...
aws
Search [Alt+S]
United States (Ohio)
SauloPJ (8713-4688-4445)
SauloPJ

ubuntu@ip-172-31-40-145:~$ sudo systemctl start postgresql
ubuntu@ip-172-31-40-145:~$ sudo systemctl enable postgresql
Synchronizing state of postgresql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable postgresql
ubuntu@ip-172-31-40-145:~$ sudo systemctl status postgresql

● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) Since Sun 2026-03-22 17:36:27 UTC; 1min 3s ago
   Main PID: 9444 (code=exited, status=0/SUCCESS)
   CPU: 2ms

Mar 22 17:36:27 ip-172-31-40-145 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Mar 22 17:36:27 ip-172-31-40-145 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
ubuntu@ip-172-31-40-145:~$
```

i-026d329df00285aa0 (fecap-instance-ubuntu)
PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145


```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06o...
AWS [Alt+S] Ask Amazon Q
Novo Chrome disponível
SauloPJ (8713-4688-4443)
SauloPJ

(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ alembic upgrade head
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
INFO [alembic.runtime.migration] Running upgrade -> e680c939651c, create users table
(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ sudo -u postgres psql -d empathic_leaders -c "\dt"
List of relations
Schema | Name | Type | Owner
-----|-----|-----|-----
public | alembic_version | table | postgres
public | users | table | postgres
(2 rows)

(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ uvicorn main:app --host 0.0.0.0 --port 8001
INFO: Started server process [10738]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
```

i-026d329df00285aa0 (fecap-instance-ubuntu)

PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145

```
us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-026d329df00285aa06o...
AWS [Alt+S] Ask Amazon Q
Novo Chrome disponível
SauloPJ (8713-4688-4443)
SauloPJ

(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ uvicorn main:app --host 0.0.0.0 --port 8001
INFO: Started server process [10824]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
INFO: 189.78.243.79:60802 - "POST /api/auth/register HTTP/1.1" 200 OK
INFO: 189.78.243.79:54218 - "OPTIONS /api/users/me/role HTTP/1.1" 200 OK
INFO: 189.78.243.79:54218 - "PUT /api/users/me/role HTTP/1.1" 200 OK
ACINFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [10824]
AC
(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ ^C
(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$ sudo -u postgres psql -d empathic_leaders -c "SELECT name, email, role FROM users;"
name | email | role
-----|-----|-----
Teste Dois | teste@gmail.com | operator
(1 row)

(venv) ubuntu@ip-172-31-40-145:~/fecap-projects/Projeto10/src/Entrega1/backend/backend-api$
```

i-026d329df00285aa0 (fecap-instance-ubuntu)

PublicIPs: 3.17.60.86 PrivateIPs: 172.31.40.145

6. Conclusão

A configuração do ambiente de computação em nuvem foi realizada com sucesso, cumprindo todos os requisitos propostos na atividade. Foi possível criar e acessar uma instância EC2 na AWS, instalar e configurar os principais softwares necessários (Git, Python 3 e PostgreSQL), clonar um repositório privado do GitHub e executar uma aplicação backend completa com banco de dados relacional em ambiente de nuvem.

A experiência proporcionou um aprendizado prático valioso sobre administração de servidores Linux, gerenciamento de serviços com `systemctl`, criação de ambientes virtuais Python, execução de migrations de banco de dados e operação de servidores web em produção. Além disso, o uso do terminal Linux no ambiente EC2 permitiu o desenvolvimento de habilidades essenciais para a área de infraestrutura e DevOps.

Os desafios encontrados ao longo do processo, como a configuração manual do arquivo `.env`, a execução das migrations e a resolução de erros do banco de dados, foram superados com base na leitura da documentação oficial e na aplicação de boas práticas de desenvolvimento. O resultado final comprova que é possível deployar uma aplicação Python completa em infra estrutura de nuvem pública utilizando recursos de baixo custo, como o tier gratuito da AWS.